

# From-Scratch Replication: Magnetostriction as a Probe of Octupolar Order under a [111] Field (Patri *et al.*, arXiv:1901.00012, 2018)

TEXTURES-100 replication (Hermes subagent)

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## 1 Model and System

The paper treats the Pr-based cage compounds  $\text{Pr}(\text{Ti}, \text{V}, \text{Ir})_2(\text{Al}, \text{Zn})_{20}$  (illustratively  $\text{PrV}_2\text{Al}_{20}$ ). The  $\text{Pr}^{3+} 4f^2$  ions on the diamond lattice (space group  $Fd\bar{3}m$ , local  $T_d$  symmetry) host a non-Kramers  $\Gamma_{3g}$  ground doublet, well separated ( $\gtrsim 50$  K) from excited states. The doublet carries *no* magnetic dipole moment; its only low-energy degrees of freedom are the quadrupoles  $O_{20}, O_{22}$  (time-reversal even) and the octupole  $T_{xyz} = \frac{\sqrt{15}}{6} \overline{J_x J_y J_z}$  (time-reversal odd,  $\Gamma_{2u}$ ). These map onto a pseudospin- $\frac{1}{2}$ :

$$\tau^x = -\frac{1}{4}O_{22}, \quad \tau^y = -\frac{1}{4}O_{20}, \quad \tau^z = \frac{1}{3\sqrt{5}}T_{xyz}. \quad (1)$$

Order parameters: AFQ  $\tilde{\varphi}$ , FQ  $\varphi$  (parasitic), and the ferro-octupole (FO)  $m \equiv \langle \tau_A^z + \tau_B^z \rangle$ .

## 2 Method (built from scratch)

Symmetry-based Landau theory plus a strain minimization, reusing the shared kernel `ollie_multipolar_stevens...` (Stevens operators, thermal susceptibility) for the single-ion multipole construction (provenance credit). Three independent computations:

1. **(A) Pseudospin construction.** Rebuild the  $\Gamma_3$  doublet from the  $J = 4$  multiplet (Eq. 2 of paper), form  $|\uparrow\rangle, |\downarrow\rangle$  (Eq. 4), and verify  $\tau^z \propto T_{xyz}$  maps onto a genuine pseudospin- $\frac{1}{2}$  operator. Also compute the single-ion octupole susceptibility  $\chi_{T_{xyz}}(T)$ .

2. **(B) Strain minimization (the headline).** With the cubic elastic energy

$$F_{\text{lat}} = \frac{c_{11}}{2}(\varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2) + \frac{c_{44}}{2}(\varepsilon_{xy}^2 + \varepsilon_{yz}^2 + \varepsilon_{xz}^2) + c_{12}(\varepsilon_{xx}\varepsilon_{yy} + \varepsilon_{yy}\varepsilon_{zz} + \varepsilon_{zz}\varepsilon_{xx}) \quad (2)$$

and the field-induced octupole-strain coupling  $\Delta F = -g_O m(\varepsilon_{yz}h_x + \varepsilon_{xz}h_y + \varepsilon_{xy}h_z)$ , minimize  $F_{\text{lat}} + \Delta F$  over the full strain tensor. This gives  $\varepsilon_{xy} = (g_O/c_{44})m h_z$  (cyclically), diagonal strains vanish, and for  $\mathbf{h} \parallel [111]$  ( $h_i = h/\sqrt{3}$ ),

$$\left( \frac{\Delta L}{L} \right)_{[111]} = \frac{\varepsilon_{xy} + \varepsilon_{yz} + \varepsilon_{xz}}{3} = \frac{1}{\sqrt{3}} \frac{g_O}{c_{44}} m h. \quad (3)$$

3. **(C) Hysteresis.** Minimize the FO Landau potential  $F_m = \frac{t_m}{2}m^2 + u_m m^4 - b(h_x h_y h_z) m$  (the cubic-in- $h$ ,  $\sim h_x h_y h_z \tau^z$  drive) on up/down field sweeps below  $T_O$  ( $t_m < 0$ ) and measure the loop width.

### 3 Results

- **(A)** Doublet orthonormal (overlaps  $10^{-16}$ ).  $\tau^z$  from  $T_{xyz}$  matches  $\frac{1}{2}\sigma_z$  with residual  $4 \times 10^{-17}$ ;  $\tau^x$  from  $O_{22}$  residual  $6 \times 10^{-17}$ . Octupole susceptibility is Curie-like:  $\chi_{T_{xyz}} = 220$  at  $T = 0.3$  vs 22 at  $T = 3.0$  (ratio exactly  $1/T$ ), supporting a ferro-octupolar instability at low  $T$ .
- **(B)** Log-log exponent of  $(\Delta L/L)_{[111]}$  vs  $h$  is 1.0000 (*exactly linear in  $h$* ). Coefficient vs  $m$ :  $R^2 = 1.000$  (*exactly  $\propto m$* ). Coefficient magnitude  $A/[(g_O/c_{44})m] = 0.57735 = 1/\sqrt{3}$  — exactly the  $[111]$  geometric projection.
- **(C)** Below  $T_O$  the octupole  $m(h)$  traces a hysteresis loop of width 1.00 (in units of the spontaneous  $m$ ), with a finite coercive field; the  $[111]$  length change inherits this hysteresis.

### 4 Comparison to the Claim

| Claim                           | Paper          | This work                      |
|---------------------------------|----------------|--------------------------------|
| $(\Delta L/L)_{[111]} \sim h^1$ | linear-in- $h$ | exponent = 1.0000 ✓            |
| coeff. $\propto m$ (FO)         | yes            | $R^2 = 1.000$ ✓                |
| coeff. $\propto g_O/c_{44}$     | yes            | ratio = $1/\sqrt{3}$ (geom.) ✓ |
| hysteresis below $T_O$          | yes            | loop width = 1.0 ✓             |

All four sub-claims of the headline are reproduced analytically/numerically. **Verdict: REPLICATED.**

### 5 Critique

The result is a symmetry-dictated identity rather than a material-specific number: given the  $T_d$ -allowed couplings, the linear-in- $h$  scaling and the  $\propto m$  proportionality follow *exactly* and are essentially parameter-free (only the geometric  $1/\sqrt{3}$  enters for  $[111]$ ). Thus the replication confirms the paper’s *internal logic* strongly, but does not independently validate (i) the microscopic value of  $g_O$  (needs the second/third order perturbation theory in  $\mathbf{h} \cdot \mathbf{J}$  and CEF gaps  $\Delta(\Gamma_4), \Delta(\Gamma_5)$ ), nor (ii) the actual value of  $T_O \approx 0.65$  K or the loop shape in  $\text{PrV}_2\text{Al}_{20}$ , which require the full temperature-dependent coupled minimization of  $F_{\tilde{\varphi}, \varphi, m}$  and experimental  $c_{44}, g_O$ . Our hysteresis is a demonstrator (single-well FO potential with cubic drive), not a fit to data. See `failure_analysis.md`.

### 6 Provenance

Single-ion angular-momentum matrices, Stevens multipole operators ( $O_{20}, O_{22}, T_{xyz}$ ), and the thermal (fluctuation) susceptibility were reused from `ollie_multipolar_stevens_landau_kernel.py` (Ollie’s shared TEXTURES-100 multipolar kernel). All Landau/strain minimization and scaling analysis are original to this replication.